

CS621/CSL622

Quantum Computing For Computer Scientists

Quantum Circuits and Protocols

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Partial measurements

Problem Definition

We have a system consisting of two or more qubits and we only measure one of them.

Example (2-qubit system)

- What is the difference between the case where **both** qubits are measured and when **only one** is measured?
 - Probabilities of measurement-outcomes of first qubit is *same* irrespective of whether second qubit was measured or not.
 - However, the **superposition** of the system after the measurement may be *different*

Both Measured

Resulting superposition is one of $|00\rangle$, $|01\rangle$, $|10\rangle$, or $|11\rangle$

One Measured

May lead to a different superposition

- Initial superposition: $|\psi\rangle = \frac{1}{2} |00\rangle - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$

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Measuring 0

Measuring 1

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Measuring 0

- Consistent terms

$$|\psi\rangle = \frac{1}{2} |00\rangle - \cancel{\frac{i}{2} |10\rangle} + \cancel{\frac{1}{\sqrt{2}} |11\rangle}$$

Measuring 1

- Consistent terms

$$|\psi\rangle = \cancel{\frac{1}{2} |00\rangle} - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

- Initial superposition: $|\psi\rangle = \frac{1}{2} |00\rangle - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$

Measuring 0

- Consistent terms

$$|\psi\rangle = \frac{1}{2} |00\rangle - \cancel{\frac{i}{2} |10\rangle} + \cancel{\frac{1}{\sqrt{2}} |11\rangle}$$

- Outcome 0 Prob.

$$\left\| \frac{1}{2} |00\rangle \right\|^2 = \frac{1}{4}$$

Measuring 1

- Consistent terms

$$|\psi\rangle = \cancel{\frac{1}{2} |00\rangle} - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

- Outcome 1 Prob.

$$\left\| -\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle \right\|^2 = \left| \frac{-i}{2} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{3}{4}$$

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- Consistent terms

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- Consistent terms

$$|\psi\rangle = \cancel{\frac{1}{2} |00\rangle} - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

- Outcome 1 Prob.

$$\left\| -\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle \right\|^2 = \left| \frac{-i}{2} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{3}{4}$$

- The state of the two qubits after the measurement **conditioned** on the measurement outcome **being 0 or 1** is the vector itself **renormalized** respectively.

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Measuring 0

- Consistent terms

$$|\psi\rangle = \frac{1}{2} |00\rangle - \cancel{\frac{i}{2} |10\rangle} + \cancel{\frac{1}{\sqrt{2}} |11\rangle}$$

- Outcome 0 Prob.

$$\left\| \frac{1}{2} |00\rangle \right\|^2 = \frac{1}{4}$$

- Renormalization

$$\frac{\frac{1}{2} |00\rangle}{\left\| \frac{1}{2} |00\rangle \right\|} = |00\rangle$$

Measuring 1

- Consistent terms

$$|\psi\rangle = \cancel{\frac{1}{2} |00\rangle} - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

- Outcome 1 Prob.

$$\left\| -\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle \right\|^2 = \left| \frac{-i}{2} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{3}{4}$$

- Renormalization

$$\frac{-\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle}{\left\| -\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle \right\|} = -\frac{i}{\sqrt{3}} |10\rangle + \sqrt{\frac{2}{3}} |11\rangle$$

- The state of the two qubits after the measurement **conditioned** on the measurement outcome **being 0 or 1** is the vector itself **renormalized** respectively.

- Initial superposition: $|\psi\rangle = \frac{1}{2} |00\rangle - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$

Measuring 0

- Consistent terms

$$|\psi\rangle = \frac{1}{2} |00\rangle - \cancel{\frac{i}{2} |10\rangle} + \cancel{\frac{1}{\sqrt{2}} |11\rangle}$$

- Outcome 0 Prob.

$$\left\| \frac{1}{2} |00\rangle \right\|^2 = \frac{1}{4}$$

- Renormalization

$$\frac{\frac{1}{2} |00\rangle}{\left\| \frac{1}{2} |00\rangle \right\|} = |00\rangle$$

Measuring 1

- Consistent terms

$$|\psi\rangle = \cancel{\frac{1}{2} |00\rangle} - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

- Outcome 1 Prob.

$$\left\| -\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle \right\|^2 = \left| \frac{-i}{2} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{3}{4}$$

- Renormalization

$$\frac{-\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle}{\left\| -\frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle \right\|} = -\frac{i}{\sqrt{3}} |10\rangle + \sqrt{\frac{2}{3}} |11\rangle$$

- Thus, conditioned on the measurement outcome, the state of the two qubits becomes one of the following:

$$|00\rangle \quad \text{or,} \quad -\frac{i}{\sqrt{3}} |10\rangle + \sqrt{\frac{2}{3}} |11\rangle$$

- Find the state of the two qubits if the second qubit is measured.

$$|\psi\rangle = \frac{1}{2} |00\rangle - \frac{i}{2} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

Measure Measure Measure

- Initial State:

$$|\psi\rangle = \frac{1}{2} |000\rangle + \frac{1}{2} |100\rangle + \frac{1}{2} |101\rangle - \frac{1}{2} |111\rangle$$

Measure
Measure
Measure
Measure

- Partial measurement: Qubit-3
- What are the probabilities of the two possible measurement outcomes and what are the resulting superpositions of the three qubits for each case?
- We write:

$$|\psi\rangle = \left(\frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right) |0\rangle + \left(\frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) |1\rangle$$

- Inconsistent Terms

$$|\psi\rangle = \left(\frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right) |0\rangle + \left(\frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) |1\rangle$$

- $\Pr[M(\text{Qubit} - 3) = 0]$

$$\left\| \frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right\|^2 = \frac{1}{2}$$

- Resulting Superposition:

$$\frac{\left(\frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right) |0\rangle}{\left\| \frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right\|} = \sqrt{2} \left(\frac{1}{2} |00\rangle + \frac{1}{2} |10\rangle \right) |0\rangle$$

$$= \frac{1}{\sqrt{2}} |000\rangle + \frac{1}{\sqrt{2}} |100\rangle$$

Renormalization

- Inconsistent Terms

$$|\psi\rangle = \left(\frac{1}{2} |00\rangle - \frac{1}{2} |10\rangle \right) |0\rangle + \left(\frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) |1\rangle$$

- $\Pr[M(\text{Qubit} - 3) = 1]$

$$\left\| \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right\|^2 = \frac{1}{2}$$

- Resulting Superposition:

$$\frac{\left(\frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) |1\rangle}{\left\| \frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right\|} = \sqrt{2} \left(\frac{1}{2} |10\rangle - \frac{1}{2} |11\rangle \right) |1\rangle$$

$$= \frac{1}{\sqrt{2}} |101\rangle - \frac{1}{\sqrt{2}} |111\rangle$$

Renormalization

1. Quantum Computing for Computer Scientists, by Noson S. Yanofsky, Mirco A. Mannucci
2. Quantum Computing Explained, David McMahon. John Wiley & Sons
3. Lecture Notes on Quantum Computation, John Watrous, University of Calgary
 - <https://cs.uwaterloo.ca/~watrous/QC-notes/>